



WORD FREQUENCY & AUTHORSHIP

Mini Project

BY: GURURAJ KHOT

SOURCES

(Project Gutenberg)

"Great Expectations" by **Charles Dickens**

"David Copperfield" by **Charles Dickens**

"The Adventures of Sherlock Holmes" by **Arthur Conan Doyle**

"The Hound of the Baskervilles" by **Arthur Conan Doyle**

QUESTIONS

Can word frequency tell two authors apart?

Do two books by one author look more alike than books by different authors?

Hard mode: both authors are Victorian, British, and all four novels are first person — so era and voice cannot explain any difference.

APPROACH

1. Download 4 books, strip the Gutenberg header & footer
2. Lowercase, drop punctuation, remove stop words
3. Normalize to a rate per 10,000 words
4. Rank words by the Dickens / Doyle ratio
5. Test with a binomial z-test — within AND across authors

P R O B L E M S

- **Typographic apostrophes split every contraction***
- Character names swamped the results (Pip, Holmes)
- Books differ ~6x in length, so raw counts mislead
- Top words reflect genre, not writing style

* These files use " ' " not " ' ", so "ain't" became "ain" + "t" and "Em'ly" became "em" + "ly". Those fragments ranked as Dickens's most distinctive words until I normalized the apostrophe.

Charles Dickens

1. said
2. mr
3. more
4. miss
5. know
6. time
7. little
8. upon
9. about
10. never

R E S U L T S

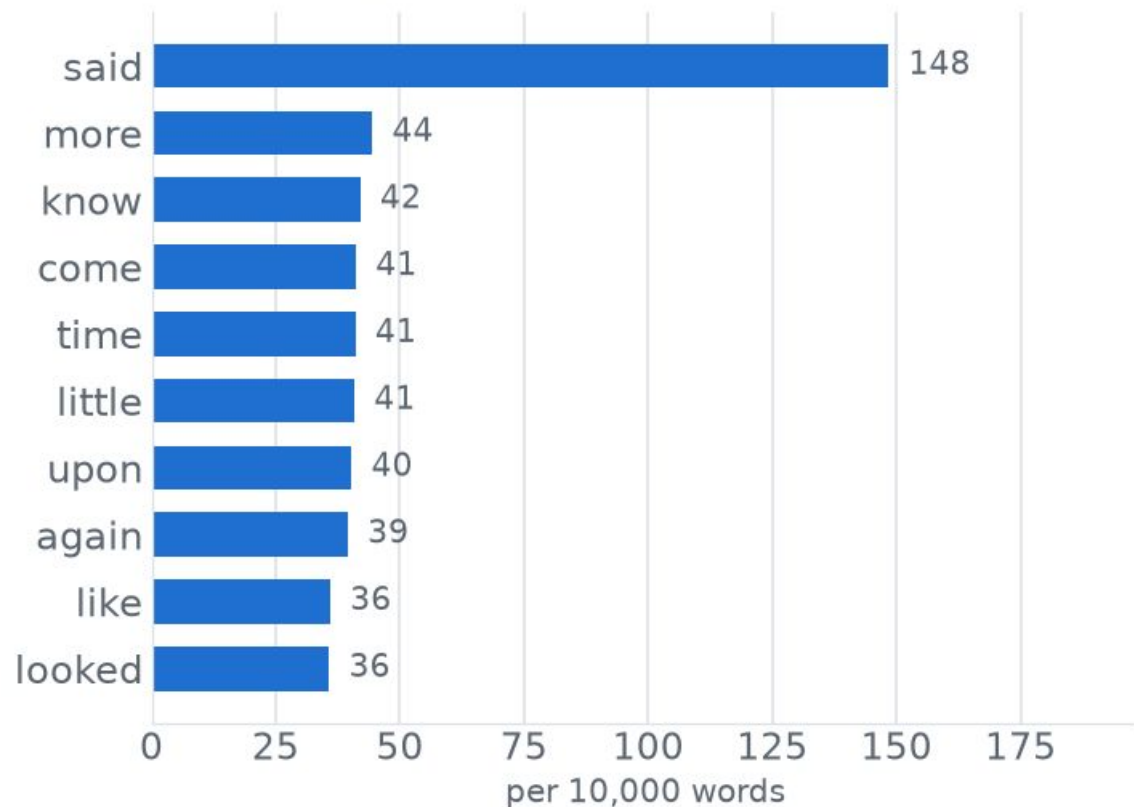
Arthur Conan Doyle

1. said
2. upon
3. man
4. more
5. know
6. over
7. see
8. well
9. about
10. may

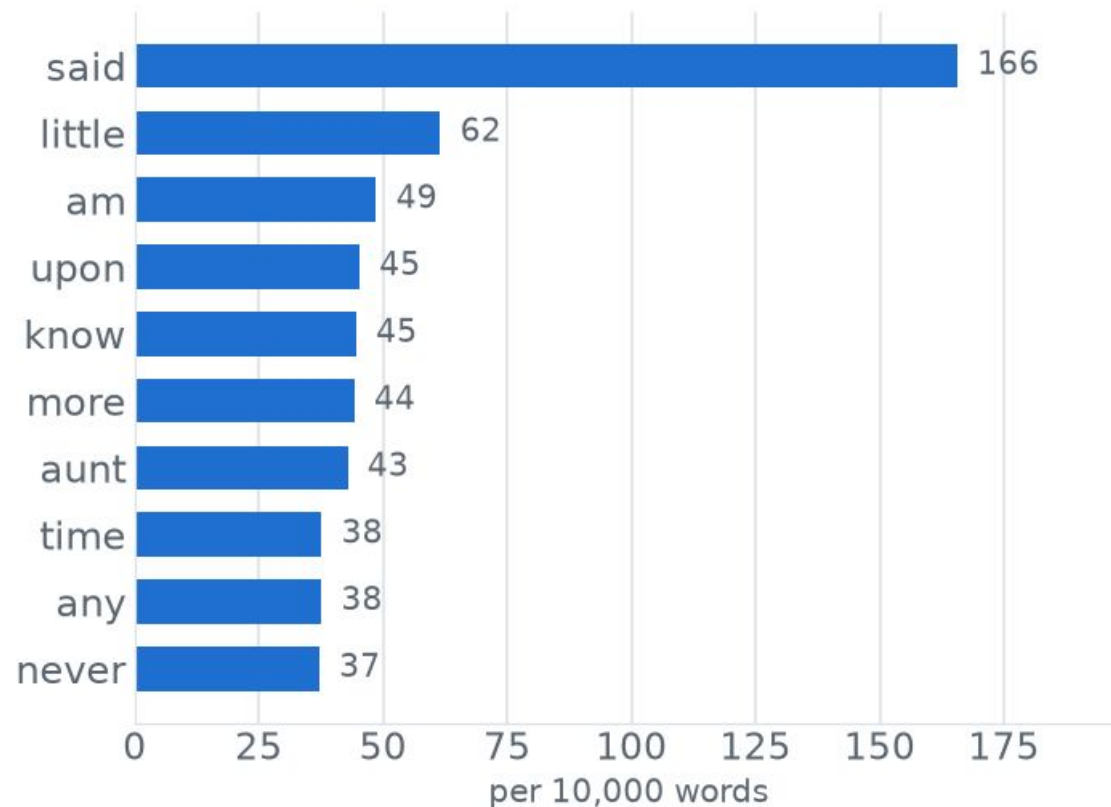
Top words after stop words and character names are removed

Charles Dickens — his two novels

Great Expectations

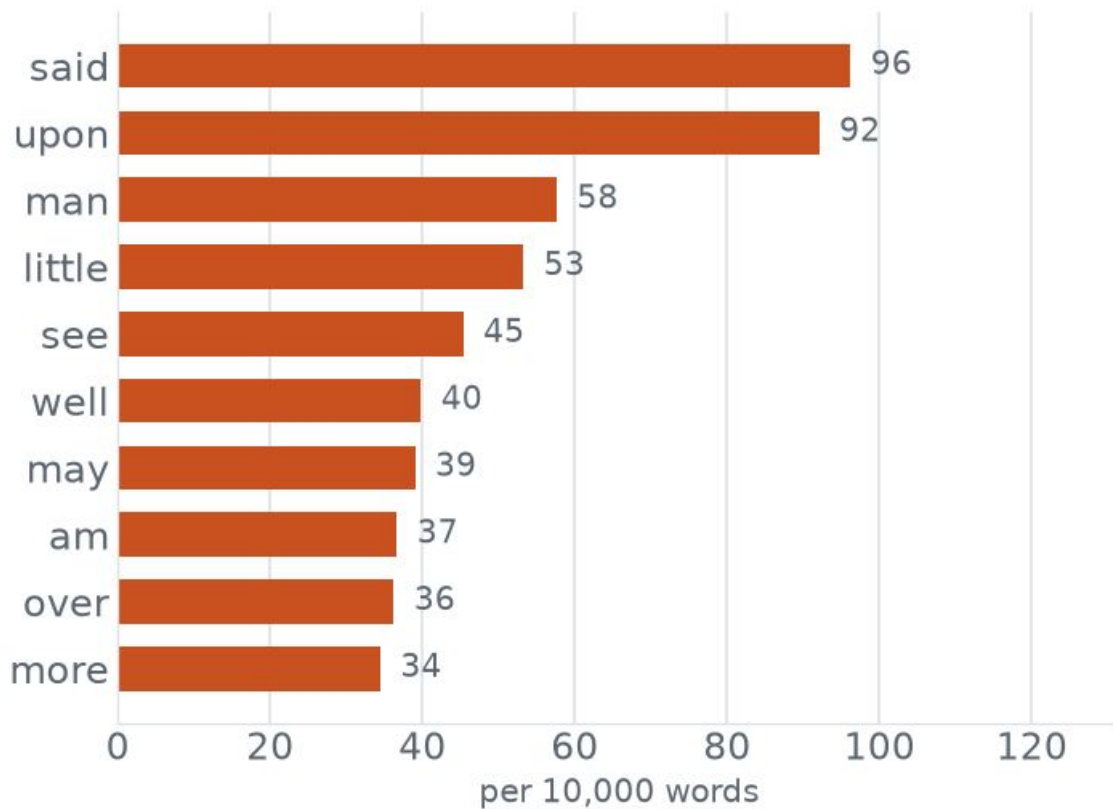


David Copperfield

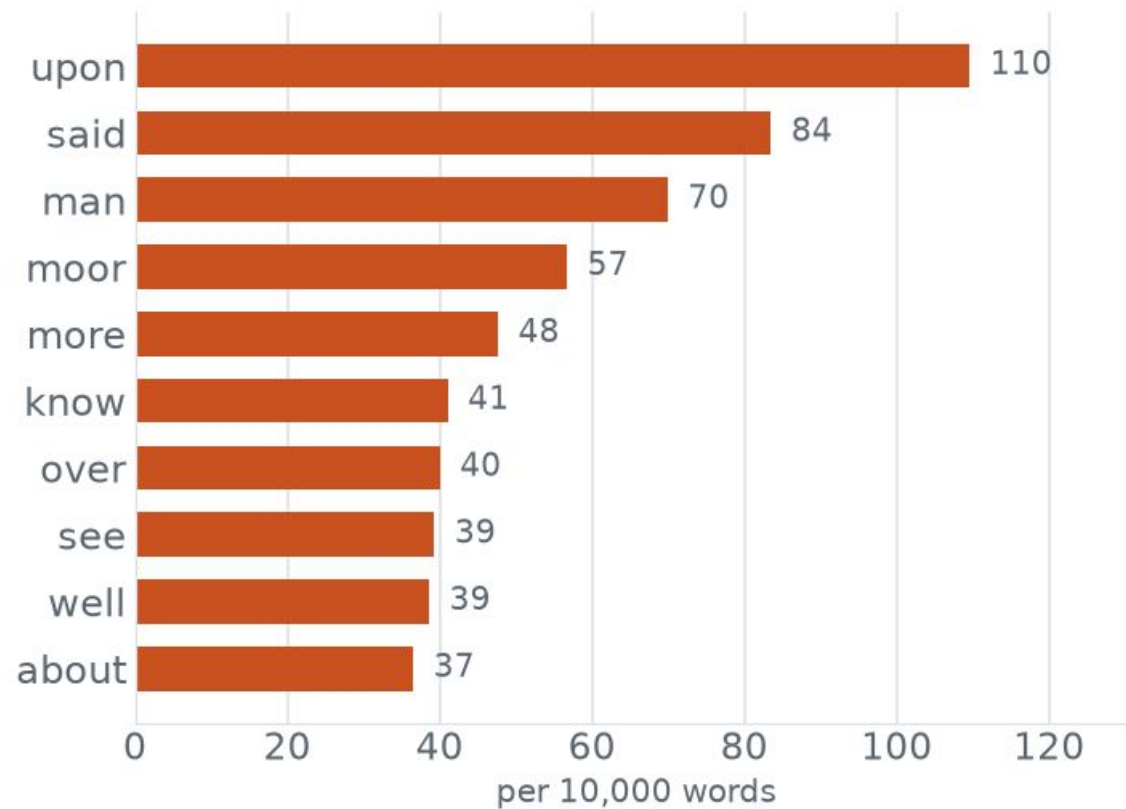


Arthur Conan Doyle — his two novels

Sherlock Holmes

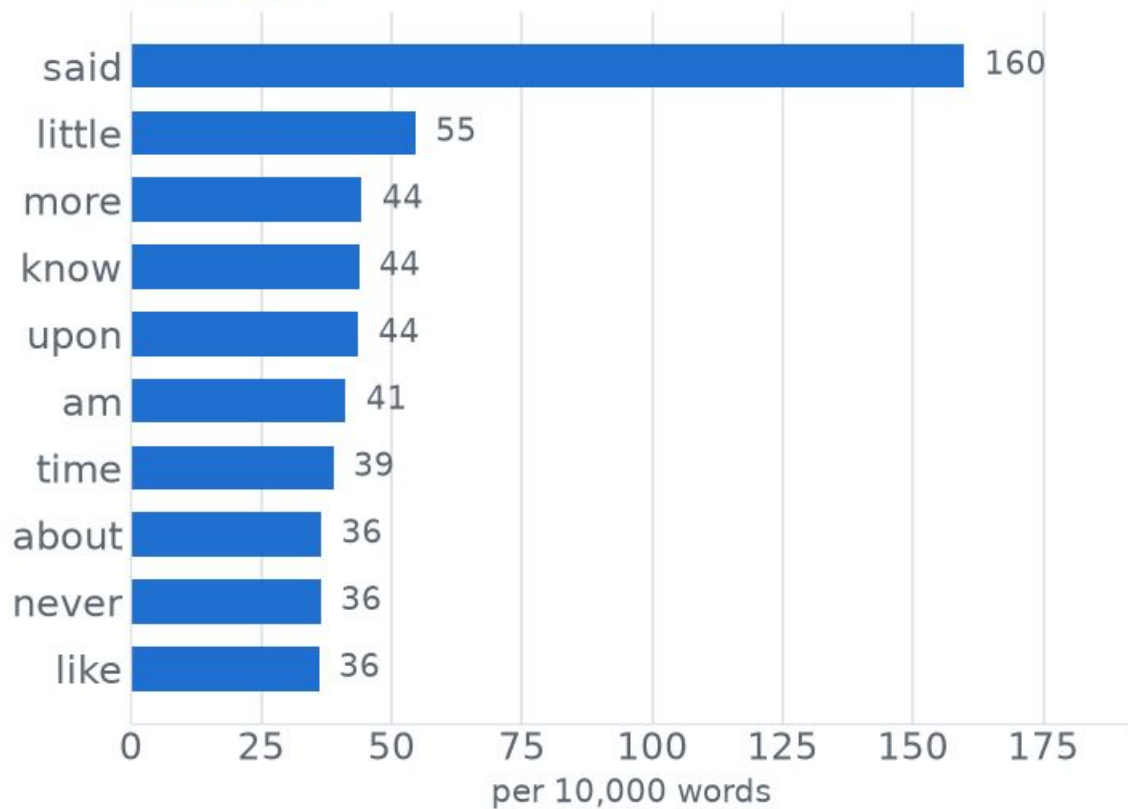


Hound of the Baskervilles

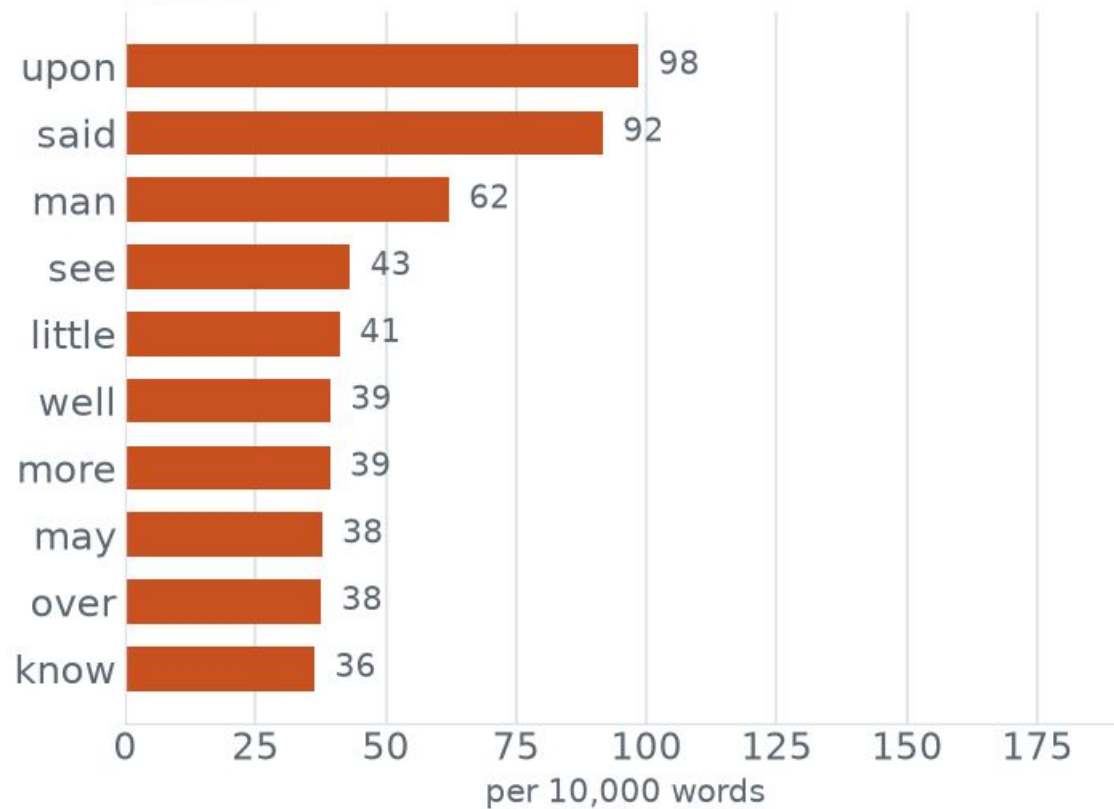


Two authors, pooled — same words, different weights

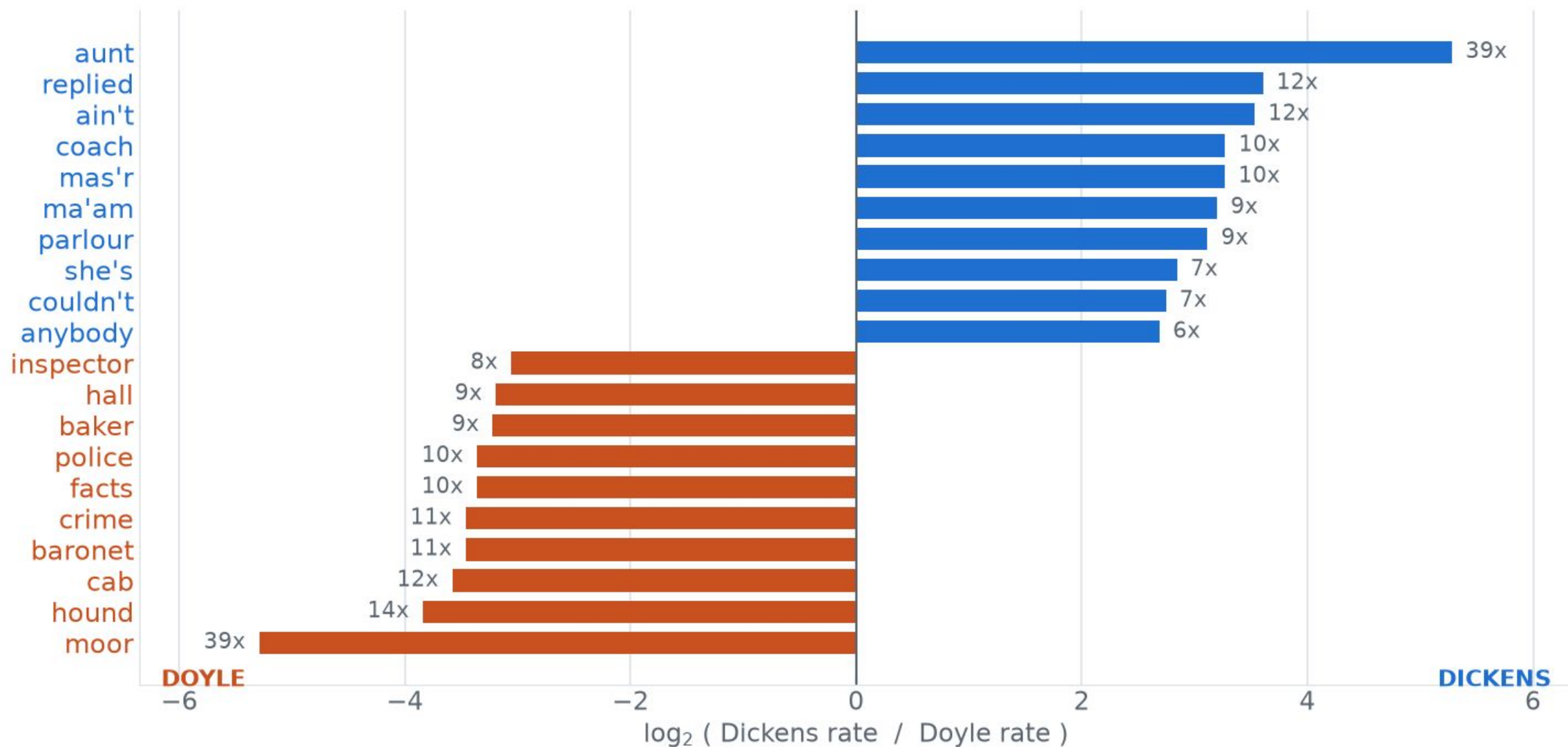
Dickens



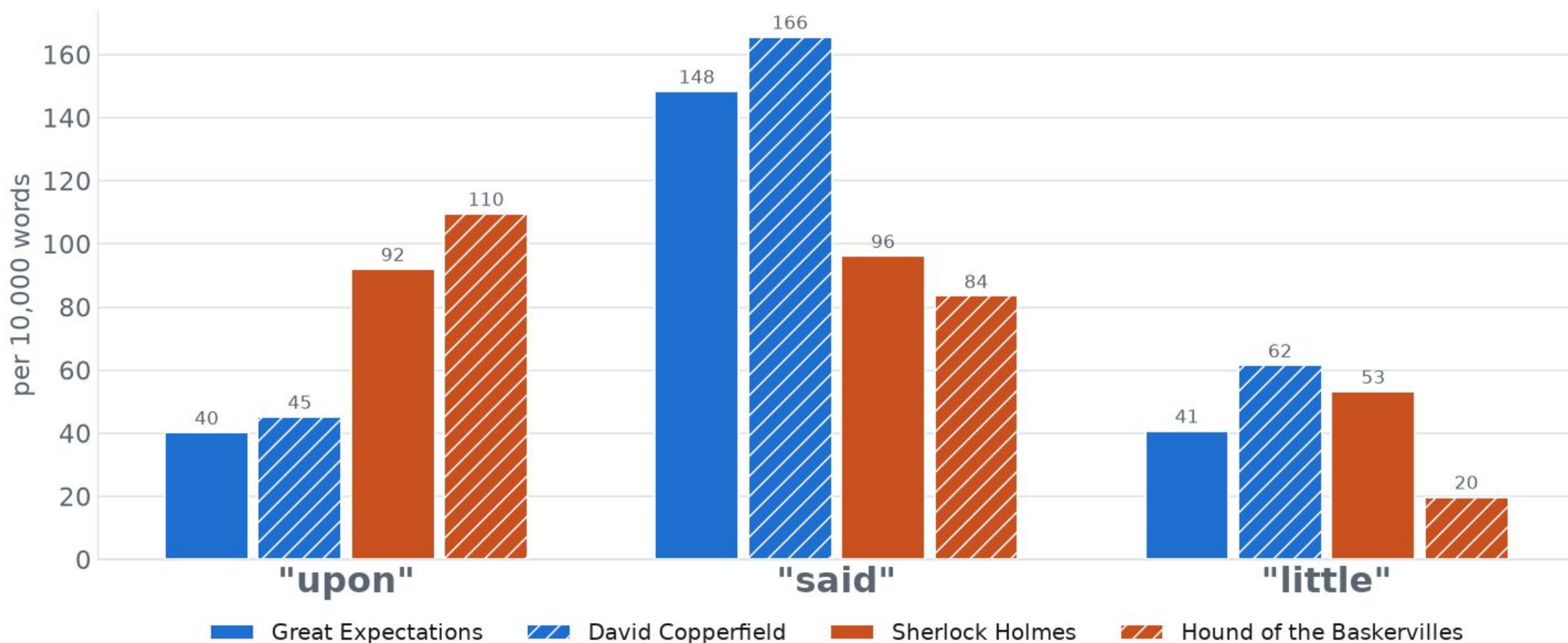
Doyle



Words one author uses and the other does not



Two markers that work, one that fails



NOTES

- **Each author is consistent with himself** — the two Dickens books share 13 of their top 20 words, Doyle's share 12
- **Authors differ by weight, not vocabulary** — "upon" is 44 per 10k for Dickens and 98 for Doyle
- **"little" is significant inside BOTH authors** — a big p-value difference that is useless for authorship
- **Median $|z|$ is 2.13 within an author vs 8.56 across** — about a fourfold gap

NEW IDEAS

- Are the stop words I threw away the real signal?
- Yes — "from" separates the authors 14x better than either differs from himself
- Would same-genre books remove the crime-vs-aunts effect?
- Could this scale to a classifier over many authors?
- Word counting is embarrassingly parallel — map over books, reduce the counters

The End!